



ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE
DIGITAL HUMANITIES LABORATORY (DHLAB)

THE SOCIAL LIFE OF IMAGES :

Using Social Metadata to Enhance AI Generated Descriptions of Digital
Image Archives, Visual place recognition for Flickr Commons photographs
via Mapillary feature matching

BACHELOR SEMESTER PROJECT

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ABSTRACT

The *Social Life of Images* programme at the EPFL Digital Humanities Laboratory aims to enrich historical photographs from the Flickr Commons with precise spatial, temporal, and descriptive context. A precondition for that goal is reliable photo-level geolocation: institutional GPS coordinates attached to Flickr Commons records are often coarse, shoot-level, or point to the archive's address rather than the depicted site, which limits any downstream module that assumes coordinates identify where the picture was taken. This report presents a verification-and-refinement layer that consumes building-positive photographs from the filtering stage and re-anchors each one to a modern street-level panorama drawn from *Mapillary*.

The method combines three components: a candidate sampler that draws viewpoint-diverse Mapillary images around each query through a local tile grid and a Gaussian-weighted bbox sampler; LoFTR detector-free local feature matching [1] between the historical photograph and each candidate; and RANSAC geometric verification with the MAGSAC robust estimator [2, 3], with the resulting inlier count used both to rank candidates and as an uncertainty proxy in the spirit of Sferrazza et al. [4]. The matcher is exposed as a single function whose output columns match the production database schema, so the same code serves a local CSV development loop and the database-backed trainer maintained by Nathanael Lambert.

This report documents the design iterations behind the final pipeline, reports results statistics and runtime, and discusses the limitations Mapillary coverage gaps, historical/modern domain shift, and more.

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CHAPTER 1

INTRODUCTION

1.1 CONTEXT

The *Social Life of Images* project at the EPFL Digital Humanities Laboratory (DHLAB) investigates how large heritage photograph corporation principally the Flickr Commons, can be filtered, classified, and re-organised to support spatially and temporally grounded research. A long-term objective within that programme is a contextual enrichment model: given a heritage photograph, infer when and where it was taken, retrieve other photographs of architecturally similar structures, and aggregate the surrounding social metadata to produce richer descriptions than any single catalogue record allows.

That model rests on several prerequisite layers. A first layer, filters the corpus to photographs that actually depict architecture and groups confirmed positives into geographic neighbourhoods using DBSCAN on Flickr-supplied coordinates. The present report contributes the next layer in that chain: **per-photo geolocation refinement and verification**. Given a building-positive photograph and its (often unreliable) Flickr coordinate, the goal is to locate a corresponding modern street-level panorama in Mapillary, confirm the visual correspondence through feature matching, and emit a refined coordinate together with an interpretable confidence score.

The work is delivered as a single matching function whose output columns match those of the shared `machine_learning_photo` table, allowing the matcher to be consumed both by a local CSV runner used during development and by the production database trainer maintained by Nathanael Lambert without modification. The bridge between local experimentation and production deployment is the function’s output schema rather than a translation layer.

1.2 MOTIVATION

Flickr Commons coordinates are heterogeneous in quality. Some institutions geotag every image to the archive’s physical address; others place all photographs from one accession at a single shoot-level point; a minority encode genuine per-photo GPS. For an enrichment model that intends to retrieve “other photos of the same building” or “other photos near here”, this matters: a 150 m error is the difference between matching a cathedral and matching the city hall across the square. Aggregate funnel statistics from the filtering stage confirm the scale of the problem, coordinates exist for only 3.3% of metadata rows, and even among those, clustering produces many groups whose members are co-located but do not depict the same structure.

Modern open street-level imagery offers a way to re-anchor each photograph independently of the catalogue. Mapillary in particular provides dense, openly licensed coverage in most urban centres and an

API that exposes both image metadata and vector tiles. The strategy adopted here is therefore: for each building-positive query, retrieve a compact set of nearby Mapillary panoramas, perform pairwise feature matching between the archive image and each candidate, and select the candidate that yields the most geometrically consistent correspondences after RANSAC verification.

Two design decisions follow from the heritage setting. First, the matcher must handle severe domain shift black-and-white film, hand-drawn artefacts on the negative, century-scale changes to façades and surroundings, and substantial viewpoint differences from modern dashcam-style Mapillary captures. This argues for LoFTR’s detector-free coarse-to-fine attention-based matching [1] over detector-based pipelines such as SuperPoint+SuperGlue, which struggle on low-texture and repetitive architectural surfaces. Second, the system must be honest about failure: many heritage photographs have no usable modern counterpart, either because the building has been demolished, because Mapillary coverage is absent at that location, or because the viewpoint is unreachable from public roads. Following Sferrazza et al. [4], the post-RANSAC inlier count is used not only to rank candidates but as a calibrated proxy for match confidence, with a documented threshold below which the system declines to commit.

1.3 CHALLENGES

The operational setting raises several practical difficulties. Mapillary coverage is uneven and partially adversarial to monument photography: dense urban roads return hundreds of near-identical car-mounted frames, while pedestrianised heritage sites and rural monuments can have no coverage at all. Naive bounding-box queries therefore over-fetch and under-cover simultaneously. LoFTR is a relatively expensive model and is the dominant cost in the pipeline, so candidate count per query must be kept small without sacrificing the chance of including the correct viewpoint. The historical–modern domain shift means even visually correct matches can produce modest inlier counts, complicating threshold calibration. Finally, the absence of pixel-level or photo-level ground truth at corpus scale limits formal evaluation to spot-checks, distribution-level statistics, and inspection of the side-by-side comparison artefacts the pipeline emits.

1.4 GOALS

The parent enrichment programme aims at full temporal and spatial contextualisation of heritage photographs. The objective assigned to this project is narrower and instrumental: build a reproducible pipeline that, for a building-positive Flickr photograph, returns either a verified Mapillary match with refined coordinates and a confidence score, or an explicit no-match decision. Concretely the work pursues five interconnected objectives.

1. **Mapillary candidate sampling.** Extract a compact, viewpoint-diverse candidate set per query through a local tile grid and a Gaussian-weighted bounding-box sampler, rather than uniform image fetching.
2. **Feature matching.** Pair each query with each candidate through LoFTR and verify correspondences with the MAGSAC variant of RANSAC.
3. **Confidence.** Derive a per-match score from the inlier count, define an operational review threshold of 15 inliers ($p_{\text{match}} \geq 0.016$), and document the production retention threshold of $p_{\text{match}} > 0.05$ applied by the database cleanup routine.
4. **Analysis artefacts.** Produce side-by-side comparison images that support manual quality review of matches and failures.

5. **Single function, two consumers.** Package the matcher behind a single function whose output columns match those of the shared `machine_learning_photo` table, so the same code serves a local CSV development loop and Nathanael Lambert’s database-backed trainer without modification.

1.5 REPORT STRUCTURE

Chapter 2 details the methods: the two execution paths sharing the same matcher, candidate sampling, LoFTR matching, RANSAC verification, and the confidence formulation. Chapter 3 reports statistics, the strong-match rate, the distribution of GPS corrections, and some worked examples. Chapter 4 interprets design iterations, limitations (including the bounded correction range imposed by the search radius), and integration outlook. Chapter 5 summarises deliverables and directions for future work.

CHAPTER 2

METHODS

2.1 OVERVIEW

The pipeline consumes building-positive photographs from the upstream filtering stage and emits, for each query, either a verified match (predicted coordinate, matched Mapillary panorama identifier, confidence score) or an explicit no-match. The four computational stages are Mapillary candidate sampling, LoFTR feature matching, RANSAC geometric verification, and ranking by inlier count. All stages share a single implementation that is consumed by two callers, as detailed in §2.2; the methods themselves are described without further reference to the calling environment in §2.3–§2.7, and the operational differences between the two callers return in §2.8.

2.2 TWO EXECUTION PATHS

The matcher described in this chapter is exposed as a single function, `find_matches(df, cache)` in `matching.py`. It accepts a pandas DataFrame of query rows and a persistent image cache, and returns the same DataFrame augmented with eight columns: `mapillary_id`, `p_match`, `mapillary_lon`, `mapillary_lat`, `mapillary_compass_angle`, `mapillary_captured_at`, `mapillary_pic_url`, and `mapillary_candidates` [5]. Every method described in §2.3–§2.6 lives behind this function. The function is the deliverable.

Two callers consume it.

Local execution (CSV). A standalone script loads a CSV of building-positive Flickr rows produced by the upstream filtering stage, calls `find_matches`, and writes the augmented DataFrame back to disk via `write_csv` in `geo_utils.py`. The 1,768-photo batch reported in Chapter 3 was produced in this mode. It is the surface used during development for prompt and threshold tuning, for inspecting the side-by-side comparison images that `save_comparison` emits, and for the distributional analysis that informs Chapter 4.

Production execution (database). The same `find_matches` function is imported by `pipeline.fast_mapillary` in the production trainer (Nathanael’s project). The trainer selects rows whose `p_match` is still `NULL` from the shared `photo/machine_learning_photo` join, calls `find_matches` one row at a time, and persists the returned columns to the database through `update_ml_photo`. Resume is implemented at the database level: the next invocation of the trainer picks up where the previous one left off because the `p_match IS NULL` filter naturally excludes rows already processed.

The bridge between the two paths is the function signature and the output column names. The matcher

does not know which caller invoked it, and neither caller needs to understand anything about LoFTR, RANSAC, or Mapillary’s API beyond what `find_matches` returns. The column names were chosen to match the columns already defined on the `machine_learning_photo` table, so the production caller does not have to translate or rename anything between the matcher’s output and the database write. This is the same design discipline applied to the building-filtering layer: a single algorithmic core with thin, equivalent adapters on either side.

2.3 MAPILLARY CANDIDATE SAMPLING

The pipeline draws Mapillary candidates around each query coordinate using two complementary helpers in `mapillary_client.py`. Both rely on Mapillary’s `/images` graph endpoint, which returns image metadata inside a bounding box, rather than on vector tiles; earlier vector-tile-based variants were tried and are discussed in Chapter 4.

LOCAL 3×3 TILE GRID WITH ANGLE BINNING (`SMART_ANGLE_CANDIDATES`). Around the query the function builds a 3×3 grid of square tiles, each of side `TILE_SIDE_KM = 0.1` km, so the total search area covers roughly 300 m × 300 m centred on the Flickr coordinate. For each of the nine tiles, one Mapillary `/images` request is issued with `limit=1000` and the fields needed downstream (`id`, `geometry`, `captured_at`, `compass_angle`, `thumb_1024_url`). Returned panoramas are bucketed by compass heading into bins of size `BIN_SIZE = 15°` (24 bins covering the full circle), and one panorama is drawn uniformly at random from each non-empty bin. The output is the concatenation across all nine tiles. The cap of one pick per heading bin per tile keeps the candidate list compact while preserving viewpoint diversity, without it, dense urban roads return dozens of near-identical car-mounted frames pointing in the same direction.

GAUSSIAN-WEIGHTED SAMPLER (`CREATE_SAMPLER + MAPILLARYSAMPLER.SAMPLE_CANDIDATES`). A complementary helper fetches a single bounding box of half-width $5 \cdot \text{st_km}$ (default `st\_km = 0.05` km, giving a 0.5 km box side) with `limit=1000`, then wraps the returned panoramas in a `MapillarySampler` object. Calling `sample_candidates` draws one image at a time, weighting each candidate by an unnormalised 2-D Gaussian PDF centred on the query,

$$w_i \propto \exp\left(-\frac{1}{2} \left(\frac{d(\text{query}, c_i)}{\sigma}\right)^2\right), \quad \sigma = \text{st_km}, \quad (2.1)$$

where $d(\cdot, \cdot)$ is the haversine distance. The weighting biases draws toward panoramas captured close to the Flickr coordinate while still admitting more distant ones. Sampling is performed via `random.choices`, which normalises the weights internally.

The two functions serve complementary purposes the angle-binned grid produces a structured, viewpoint-diverse candidate pool sized for batch matching against LoFTR, while the Gaussian sampler provides a lightweight stochastic draw biased toward the query’s reported location. The candidate counts in the output CSV in Chapter 3 reflect the combined behaviour at scale. Note that the 3×3 tile grid extends approximately 150 m from the query in each cardinal direction, so the geometric ceiling on the displacement between a query and any returned candidate is $\sqrt{2} \times 150 \approx 212$ m; this matters when interpreting the distance-correction distribution in Chapter 3, whose maximum sits just inside that bound.

2.4 LOFTR FEATURE MATCHING

Each (query, candidate) pair returned by the sampler is scored with LoFTR [1], a detector-free coarse-to-fine transformer-based matcher. Unlike detector-and-descriptor pipelines such as SuperPoint+SuperGlue,

LoFTR skips the keypoint detection stage entirely: a convolutional backbone produces feature maps at $1/8$ and $1/2$ of the input resolution, the coarse maps are passed through interleaved self- and cross-attention layers that condition each location on the global content of both images, a differentiable matching layer establishes dense correspondences at $1/8$ resolution, and selected coarse matches are refined to sub-pixel accuracy in small local windows at the fine level. The detector-free design is the property that matters here: heritage façades commonly present low-texture and repetitive regions (smooth ashlar walls, regularly spaced windows, fluted columns) where keypoint detectors fail to produce repeatable interest points across the historical–modern domain gap, and where the global receptive field of the attention mechanism continues to produce usable correspondences.

The implementation in `building_matcher.py` uses the Kornia `KF.LoFTR` wrapper loaded with the `outdoor` checkpoint, which is more appropriate for the architectural query distribution here than the indoor variant. The model is lazily instantiated once per session by `load_matcher`, moved to CUDA when available and to CPU otherwise, and reused across all pairs to amortise the construction cost. A guard variable (`load_failed`) ensures that an initial load failure is not retried on every subsequent candidate, since LoFTR loading depends on a checkpoint download from a public mirror that can fail on networks with strict certificate validation.

Inputs are prepared by `to_gray_tensor`: each PIL image is converted to grayscale, rescaled so the longest side does not exceed a fixed maximum, and then cropped so both dimensions are divisible by 8 the stride at which LoFTR’s coarse features are computed. The grayscale array is normalised to $[0, 1]$ and shaped to the $[1, 1, H, W]$ tensor that LoFTR expects, then transferred to the model’s device. The pair (t_1, t_2) is passed through the matcher inside a `torch.no_grad()` block in `compute_loftr_matches`, which returns three arrays: pixel coordinates of corresponding points in the query (`keypoints0`), pixel coordinates of corresponding points in the candidate (`keypoints1`), and a per-correspondence confidence (`confidence`). Only correspondences with confidence at least $\theta_c = 0.5$ are retained for the next stage.

2.5 RANSAC GEOMETRIC VERIFICATION

LoFTR confidence reflects appearance similarity but not geometric consistency. Two visually similar regions in unrelated images repeated window mouldings, similar paving textures, identical lamp posts can produce high-confidence correspondences that no single camera geometry could account for. The surviving correspondences are therefore passed to a geometric verification stage that asks whether they can be jointly explained by a single fundamental matrix between the two views.

The implementation in `compute_ransac_inliers` calls OpenCV’s `cv2.findFundamentalMat` with the USAC_MAGSAC robust estimator [2, 3], a reprojection threshold of $\tau = 3.0$ pixels, a stopping confidence of 0.99, and a maximum of 500 iterations. USAC_MAGSAC was chosen in preference to plain RANSAC because it marginalises over a range of noise scales rather than requiring a single user-defined inlier threshold, which makes it more stable when the correspondence set is small and noisy both common in the historical–modern matching regime here. A short-circuit at the top of the function returns zero inliers immediately when fewer than eight correspondences survived LoFTR’s confidence filter, because the seven- and eight-point algorithms used internally by the estimator require at least eight points to fit a fundamental matrix.

The output is the integer count of correspondences whose reprojection error against the estimated fundamental matrix is within τ , i.e. the number of LoFTR matches that survived geometric verification. This inlier count is the quality signal carried through the rest of the pipeline.

2.6 CANDIDATE RANKING AND CONFIDENCE

For each query the candidate with the largest inlier count is selected as the best match. The corresponding Mapillary coordinate is reported as the predicted location and the displacement from the original Flickr coordinate is computed with the haversine formula

$$d_{ij} = 2R \arcsin \sqrt{\sin^2 \frac{\Delta\phi}{2} + \cos \phi_i \cos \phi_j \sin^2 \frac{\Delta\lambda}{2}}, \quad (2.2)$$

with $R = 6,371$ km.

The raw inlier count is mapped to a bounded confidence score for storage in the `p_match` column by

$$p_{\text{match}} = \min\left(\frac{\text{inliers} + 1}{1000}, 0.999999\right). \quad (2.3)$$

The +1 in the numerator distinguishes a candidate that was geometrically verified to have zero inliers from a query that produced no candidate at all (which is recorded as a NULL `p_match` and `status = no_result`). The cap below 1.0 prevents downstream consumers from treating any single geometric verification as logically certain.

TWO THRESHOLDS. Two thresholds are documented in this report. A permissive **review threshold** of 15 RANSAC inliers, equivalent to $p_{\text{match}} \geq 0.016$ under Eq. (2.3), marks matches that are surfaced for human inspection through the side-by-side comparison images. A stricter **retention threshold** of $p_{\text{match}} > 0.05$, equivalent to roughly 49 inliers, is applied by the production database pipeline to decide which matches are persisted in `machine_learning_photo` and which are blanked back to NULL by the cleanup routine `rm_low_confidence_matches`. The two thresholds reflect different intents: the review threshold is broad enough that no plausible match is hidden from the reviewer during qualitative inspection, while the retention threshold is strict enough that only matches the matcher is confident about become part of the corpus’s permanent annotation. The choice of 15 inliers was set during development by inspecting the distribution of inlier counts across the 1,768-photo batch alongside the corresponding comparison images, and represents the smallest inlier count at which manual review consistently agreed with the matcher’s choice. The 0.05 cleanup threshold is inherited from the production pipeline and reported here for completeness.

The use of inlier counts both as a re-ranker and as an uncertainty proxy follows the argument of Sferrazza et al. [4] that geometric verification provides a more reliable confidence signal than retrieval similarity in modern visual place recognition pipelines, particularly under domain shift.

2.7 OUTPUTS

The matcher’s per-query output is a small structured record that travels through both execution paths described in §2.2. The central column is `p_match`: the bounded confidence score defined in §2.6. `p_match` is the operational ranking signal for the pipeline. It induces a total order over processed photographs and is the column that downstream consumers manual reviewers browsing the side-by-side comparison images, the production cleanup routine that retains only matches with $p_{\text{match}} > 0.05$, and any future module that wants to weight retrieval results by match confidence should read first.

The remaining columns support the ranking decision rather than competing with it. The Mapillary metadata pulled through from the matched panorama (`mapillary_id`, `mapillary_lat`, `mapillary_lon`, `mapillary_compass_angle`, `mapillary_captured_at`, `mapillary_pic_url`) describes the evidence the decision rests on; `mapillary_candidates` records how many panoramas were

considered before ranking. A `status` field carries the value `ok` when a candidate was returned by the sampler and processed through LoFTR and RANSAC, or `no_result` when the sampler returned no candidates (typically because Mapillary has no coverage in the query’s bounding box); rows with `no_result` carry NULL values for all match-dependent columns including `p_match` itself, which lets consumers distinguish a low-confidence match from an absent one without inspecting individual values.

The column names mirror the columns already defined in Nathanael’s project because the production path writes directly into them. The CSV path uses the same names for the same reason: it is the same DataFrame schema in both cases, only the destination differs.

For every row with `status = ok` the pipeline additionally renders a side-by-side comparison JPEG via `save_comparison`. The output canvas places the archive photograph on the left and the matched Mapillary thumbnail on the right at a common height of 600 pixels, separated by a 16-pixel gap and topped by a 70-pixel caption strip carrying the institution, the date taken when available, the title (truncated to 90 characters if longer), and the predicted coordinates. Output filenames combine a sanitised title with the photo identifier, which keeps comparison images recognisable when browsed in a file manager. Sorting these JPEGs by the `p_match` of the corresponding row produces a quick visual triage from highest-confidence matches at the top to weakest at the bottom, the artefact most useful for the failure-case analysis in Chapter 4 and for any threshold-tuning exercise that revisits the choice of 15 inliers.

2.8 INFRASTRUCTURE AND RESUME

LoFTR inference is the dominant cost in the pipeline. On CUDA the model processes each (query, candidate) pair in a couple of seconds; on CPU the same pair takes few minutes. The matcher is loaded once per session inside `load_matcher` and cached in module-level state, so that subsequent calls reuse the loaded weights and amortise both the checkpoint download and the move-to-device cost across the entire batch.

Image fetches, both for the archive photographs and for the Mapillary thumbnails, go through standard HTTP requests with 20- to 30-second timeouts; failures are caught at the per-image level so that a single broken URL fails the corresponding query but does not interrupt the batch. Mapillary requests authenticate using an access token loaded from a local `.env` file. The 3×3 tile grid in `smart_angle_candidates` issues nine Mapillary `/images` requests per query, which is the dominant network cost at the candidate-sampling stage.

RESUME ACROSS THE TWO PATHS. Resume across interrupted sessions differs between the two execution paths described in §2.2. Local execution writes one CSV at the end of the batch, so an interrupted local run loses any photographs processed since the last write; for the 1,768-photo batch reported in Chapter 3 this was acceptable because the batch fits in a single uninterrupted run. Production execution achieves natural resume through the database schema: the selection query that drives `fast_mapillary` filters for `p_match IS NULL`, so the next invocation of the trainer picks up exactly where the previous one left off, and each row is committed individually through `update_ml_photo` as soon as its match is computed. The same `IS NULL` filter also distinguishes “not yet processed” from “processed but no candidate found,” because the no-result case writes only `mapillary_candidates = 0` and leaves `p_match` itself unset, mirroring the CSV path’s `status = no_result`.

CHAPTER 3

RESULTS

This chapter reports results on the working batch of 1,768 building-positive photographs processed through the local execution path described in §2.2. All numbers in this chapter are computed from the output CSV produced by that run.

3.1 DATASET RESULTS

Table 3.1 summarises the cardinality of the pipeline at each stage, and Figure 3.1 shows the same counts on a logarithmic axis.

TABLE 3.1

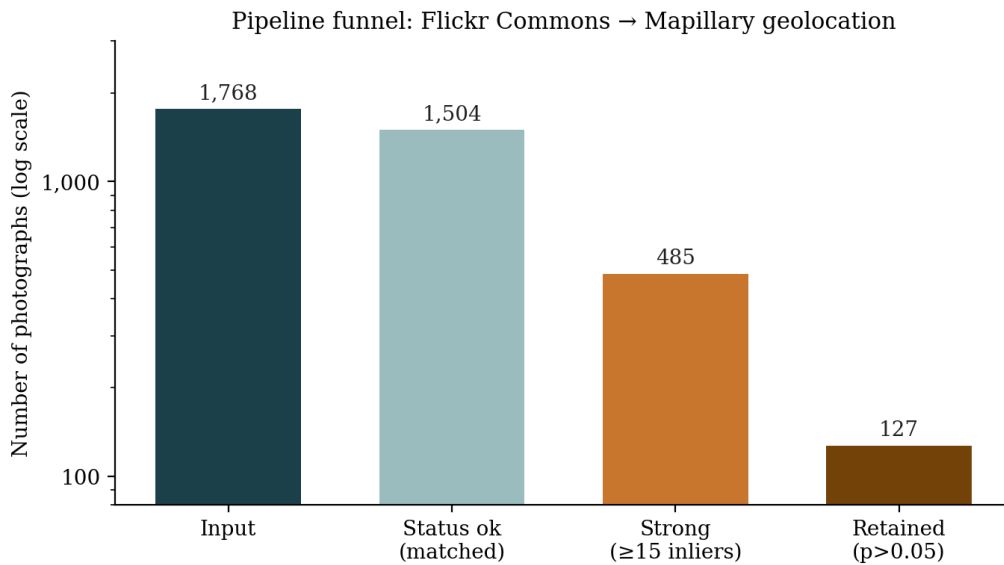
Pipeline results on the working batch. Percentages in the third column are relative to the input; percentages in the fourth column are relative to the matched subset (`status = ok`).

Stage	Count	% of input	% of matched
Input photographs	1,768	100.0%	—
<code>status = ok</code> (attempted match)	1,504	85.1%	100.0%
<code>status = no_match</code> (no candidates)	261	14.8%	—
<code>status</code> prefixed by <code>error:</code>	3	0.2%	—
Strong matches (≥ 15 inliers, $p_{\text{match}} \geq 0.016$)	485	27.4%	32.2%
Production-retained ($p_{\text{match}} > 0.05$, $\approx \geq 49$ inliers)	127	7.2%	8.4%

Two observations frame the rest of the chapter. First, the sampler has good but imperfect recall: 261 photographs (14.8%) yield no Mapillary candidate within the 3×3 tile grid, almost entirely because of absent coverage at the query coordinate. These are the cases where modern street-level imagery simply does not exist near the site the photograph depicts. Second, the gap between the matched count (1,504) and the strong-match count (485) is the central editorial decision of the pipeline: most matched rows are weakly verified the matcher returns a best candidate, but with few geometrically consistent correspondences and the report deliberately surfaces only the subset that survives the thresholds defined in §2.6.

3.2 DISTRIBUTION OF MATCH CONFIDENCE

The 1,504 matched rows have a strongly right-skewed inlier distribution. Three structural features capture its shape:

**FIGURE 3.1**

Pipeline results on a logarithmic scale. The sampler returns at least one candidate for 85.1% of input photographs; geometric verification yields a strong match (review threshold) for 27.4% of input or roughly one in three matched rows; the production retention threshold keeps 7.2% of input.

- **A spike at zero inliers (12.0% of matched rows).** These are queries where LoFTR returned fewer than the eight correspondences required by the fundamental matrix estimator, triggering the short-circuit in `compute_ransac_inliers`. They are formally matched (a best candidate was selected) but carry essentially no geometric evidence.
- **A heavy concentration in $[5, 15)$ inliers (55.7%).** This is the regime in which RANSAC fits a model but the agreement is too sparse to be conclusive the matcher’s least useful output, and the principal reason the review threshold sits at 15.
- **A right tail extending past 100 inliers (4.0%) and as far as 537.** Manual inspection of these high-inlier rows in the side-by-side comparison images consistently shows close visual agreement between the archive photograph and the Mapillary panorama, including for archive photographs taken many decades before the panorama. Two rows recorded ≥ 500 inliers and correspond to historical archive shots whose modern view appears essentially unchanged.

The median inlier count across all matched rows is 12, the 75th percentile is 17, and the 90th percentile is 41. These quantiles place the review threshold (≥ 15 inliers) just above the median so the threshold is permissive by design, intended to surface plausible matches for manual review rather than to commit to them.

3.3 GPS CORRECTION BY CONFIDENCE TIER

The headline result of the chapter is in Table 3.2: the displacement between the original Flickr coordinate and the predicted Mapillary coordinate is monotonically smaller for higher-confidence subsets.

For all matched rows the median correction is 105 m, roughly a city block but at the production retention threshold this falls to 60 m, well within the spatial scale at which an enrichment model can plausibly claim “this is a photograph of *that* building.” The 90th percentile and maximum are bounded by construction: as derived in §2.3, the 3×3 tile grid puts an upper limit of approximately 212 m on the displacement

TABLE 3.2
Distance correction quantiles by confidence tier (all values in metres).

Tier	n	Median	90th pct.	Max
All matched	1,504	105	163	210
Strong (≥ 15 inliers)	485	80	156	210
Retained ($p_{\text{match}} > 0.05$)	127	60	129	195

between a query and any returned candidate, and the observed maxima (210 m for all matched, 195 m for retained) sit just inside that bound. The report therefore cannot characterise corrections beyond a city block in either direction both because the search radius caps them, and because beyond that scale the assumption that any Mapillary panorama in the bounding box plausibly depicts the same site begins to fail. Widening the search radius is discussed as future work in Chapter 4.

3.4 PER-INSTITUTION VARIATION

The match outcome varies substantially with the contributing institution, primarily because Mapillary coverage is heterogeneous across the geographic regions different institutions document. Figure 3.2 summarises the six largest contributors together accounting for 1,423 of the 1,768 input photographs, or 80% of the batch.

Top 6 contributing institutions in the working batch (1,768-photo run)

Institution	Input	Matched	Strong (≥ 15 inl.)	Retained ($p > 0.05$)	Median p_{match}	Median dist. (m)
Cornell University Library	413	367	68	8	0.013	108
Community Archives of Belleville & Hastings Co.	391	313	64	13	0.011	116
Swedish National Heritage Board	215	200	122	53	0.019	77
National Library of Ireland on The Commons	176	141	76	19	0.016	98
State Library of Queensland Australia	163	148	29	5	0.012	116
Powerhouse Museum Collection	65	65	34	6	0.016	106

Highlighted row: Swedish National Heritage Board — outlier strong-match rate (61% vs. ~20% elsewhere), driven by dense Mapillary urban coverage.

FIGURE 3.2

Match outcomes for the six largest contributing institutions in the working batch. “Matched” counts rows with `status = ok`; “Strong” counts rows with ≥ 15 RANSAC inliers; “Retained” counts rows with $p_{\text{match}} > 0.05$. Median p_{match} and median distance correction are computed over matched rows only. The Swedish National Heritage Board row is highlighted because it is the clear outlier on every quality metric.

Cornell University Library is the single largest contributor at 413 photographs, but its match-quality distribution is among the most compressed in the table: 88.9% of its photographs are matched, yet the median p_{match} sits at 0.013 (roughly 12 inliers, immediately below the review threshold), only 18.5% of matched rows clear the review threshold, and only 2.2% are retained at the production threshold. Manual inspection of the side-by-side comparison images suggests Cornell’s archive is broad and geographically diffuse: most photographs depict less-frequently-photographed sites for which Mapillary returns candidates without producing strong geometric agreement. The strongest matches in Cornell’s contribution correspond to recognisable monuments (Vadstena Castle, the Alicante Hôtel de Ville, the Mexico City Metropolitan Cathedral, the Albert Memorial), which sit in the right tail of the distribution rather than near its centre.

The Swedish National Heritage Board contribution is the visible outlier: 200 of 215 input photographs receive a match, 122 of those (61%) clear the review threshold, and 53 (26.5%) clear the production retention threshold in each case roughly three times the rate of the other institutions in the figure, and with the lowest median displacement at 77 m. This is driven by Mapillary’s dense urban coverage in Swedish cities rather than by anything specific to the heritage photographs themselves: the same monument photographed in Stockholm and in (say) Belleville will produce more matchable candidates on the Swedish side simply because more candidates and more modern panoramas are available within the 300 m search box. This is an obvious conclusion since Mapillary was founded in Malmo in 2013 before being acquired by Meta in 2020.

The full per-institution breakdown across all 41 contributing institutions in the batch is given in Appendix A.

3.5 EXAMPLES

Find below a couple of the best results that were produced in this batch :

Stockholm, Uppland, Sweden — Swedish National Heritage Board, 1945-01-01 00:00:00
Located via Mapillary at 59.333272, 18.07683



FIGURE 3.3

Stockholm, Uppland, Sweden, Swedish National Heritage Board, located at 59.333282,18.076865 with $p_{\text{match}} = 0.538$.

Saviour Gate and Saviour Tower, Moscow, Russia — Swedish National Heritage Board, 1880-01-01 00:00:00
Located via Mapillary at 55.75319, 37.62169

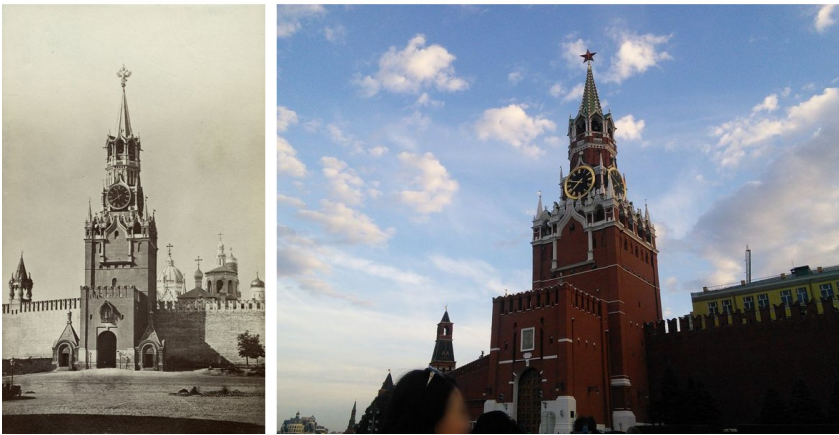


FIGURE 3.4

Saviour Gate and Saviour Tower, Moscow, Russia, located at 55.752694,37.621693 with $p_{\text{match}} = 0.134$.

"Leamington, The Town Hall" — National Science and Media Museum, 1880-01-01 00:00:00
Located via Mapillary at 52.28878, -1.53520



FIGURE 3.5

Leamington, The Town Hall, located at 52.288473,-1.5351993 with $p_{\text{match}} = 0.528$.

The building "Achtersikkel" in Ghent, Belgium — Swedish National Heritage Board, 1934-01-01 00:00:00
Located via Mapillary at 51.05380, 3.72649



FIGURE 3.6

Achtersikkel building in Ghent, Belgium, located at 51.053939,3.726457 with $p_{\text{match}} = 0.353$.

CHAPTER 4

DISCUSSION

4.1 EARLY DESIGN ITERATIONS

The pipeline arrived at its current shape through several pivots, the most consequential of which happened well before the working batch reported in Chapter 3. Documenting them clarifies why the final architecture looks the way it does and what the alternatives were.

4.1.1 FROM SIGLIP EMBEDDING SIMILARITY TO LOFTR

The first matching approach the project pursued was embedding-based. SigLIP encodes both the archive photograph and each modern Mapillary candidate into a shared 768-dimensional space, and the cosine similarity between query and candidate embeddings can in principle rank candidates by visual likeness. Two findings forced the pivot away from this approach. First, embedding similarity is a global signal: it answers “*do these two images look similar overall?*” rather than “*do they depict the same building from a compatible viewpoint?*”, and it proved unable to distinguish near-correct from incorrect matches under the historical modern domain shift, where the relevant correspondence may be a single unchanged façade against a heavily altered surrounding scene. Second, the SigLIP embedding was already being computed elsewhere in the wider *Spatial Life of Images* programme as part of the upstream filtering and dating layers, where it occupies the `sig_lip_vect_n` column of the `machine_learning_photo` table. Recomputing it inside the matcher would have duplicated work the pipeline already performs, without adding the spatial constraint that the matching stage actually requires. LoFTR replaced it because its output, dense pixel-wise correspondences that can be geometrically verified by RANSAC encodes exactly the constraint embedding similarity cannot.

4.1.2 CANDIDATE-SAMPLING ITERATIONS

4.2 LIMITATIONS AND DIFFICULTIES ENCOUNTERED

This section gathers the difficulties encountered during development some overcome, some inherent to the matching setting and the residual constraints that bound the matcher as it stands today.

CPU BOTTLENECK. The first runs of the matcher operated on CPU, where each (query, candidate) LoFTR pair takes a couple of seconds; a 200-photo development batch with 30-50 candidates per query took hours to complete, and any change to thresholds, candidate-sampling logic, or image preparation took an entire session to re-evaluate. Iterative tuning was effectively impossible in that regime. The migration to a capable environment cut the per-pair cost by roughly an order of magnitude and is what made the

1,768-photo working batch reported in Chapter 3 practically achievable in a single session overnight. The CPU code path remains as a fallback, but most of the threshold-tuning and side-by-side review work that produced the choices documented in Chapter 2 would not have been feasible without GPU acceleration.

CONFUSING EARLY-BATCH BEHAVIOUR. Early development runs surfaced two failure modes that drove the design pivots described in §4.1. The first was low overall match rate on visually obvious pairs known monuments with clear modern counterparts in Mapillary for which the matcher nonetheless returned no high-confidence result. The second was wrong matches in cases where the correct Mapillary candidate was in the candidate pool: the matcher ranked another candidate higher and committed to a visually unrelated picture. The first failure pattern indicated that the matching signal was not selective enough to identify the correct candidate even when it was present; the second indicated that the ranking signal was not discriminative enough to distinguish a near-duplicate from a generic visual match. Both observations pointed away from global embedding similarity and toward a geometrically-grounded ranker, and they are what motivated the SigLIP→LoFTR and cv2.RANSAC→USAC_MAGSAC pivots in §4.1.

SEARCH RADIUS CEILING. The 3×3 tile grid imposes a geometric ceiling of approximately 212 m on the displacement between a query and any returned candidate. As observed in Chapter 3, the maximum distance correction across the working batch sits just inside that bound (210 m for all matched rows). The pipeline therefore cannot recover from Flickr coordinates that are more than a city block away from the actual photographic viewpoint, a substantial class of errors when an institution geotags every photograph in an accession to the archive’s physical address.

COVERAGE HETEROGENEITY. Mapillary coverage across the geographic regions different institutions document is heterogeneous, and this shapes the conclusions one is allowed to draw. The Swedish National Heritage Board contribution clears the review threshold at roughly three times the rate of other institutions in the working batch, and its median distance correction is the lowest, but this reflects the density of Mapillary urban coverage in Swedish cities Mapillary was founded in Malmö and retains a dense Scandinavian footprint rather than the quality of Swedish heritage photographs. Any aggregate quality claim about the pipeline can be read as conditional on the geographic distribution of the input set.

HISTORICAL–MODERN DOMAIN SHIFT. Even visually correct matches can produce only modest inlier counts when the archive photograph is black-and-white, hand-tinted, retouched, or shows a façade altered between the time of capture and the time of the modern Mapillary panorama. The heavy concentration of matched rows in [5, 15) inliers documented in §3.2 is at least partly attributable to this effect.

4.3 OVERALL PIPELINE

The function design described in §2.2 is the integration this project delivers. `find_matches` is invoked by Nathanael Lambert’s `pipeline.fast_mapillary` with similar signature and similar output schema barring flickr metadata columns compared to the local CSV runner. The integration works on the production database today. The decision to also keep the local CSV path is operational rather than architectural it gives the development workflow somewhere to inspect threshold and prompt changes without writing to the shared database.

One asymmetry between the two paths is worth flagging. The production path commits each row individually through `update_ml_photo`, which gives natural resume across interrupted trainer sessions. The local CSV path writes the entire output in a single pass at the end of the batch, which means an interrupted local run loses any photographs processed since the last write. For the 1,768-row batch this

was acceptable because the batch fits in a single session, but at larger scales the local path would benefit from incremental row-level flushing matching the production semantics.

4.4 FUTURE WORK

Several extensions follow naturally from the system as it stands. The most substantial would replace the scalar inlier count with a structured confidence signal that combines count and spatial distribution. Each RANSAC inlier carries pixel coordinates in both the query and the candidate image; the geometry of the inlier set its compactness, its bounding-box area, the distribution of correspondences across the image plane is currently discarded. A compactness score, defined for instance as the median distance of inlier keypoints from their centroid, would let the operational threshold trade count against geometry: an acceptance rule of the form “accept if $N_{\text{inliers}} \geq f(\text{compactness})$,” with f decreasing as the inliers cluster more tightly, could recover genuine matches in the heritage regime where only a small unchanged region of the scene is shared between the archive photograph and the modern panorama. The direction of the trade-off is not obvious: standard visual place recognition pipelines typically prefer spatially spread inliers because they constrain camera geometry across the whole field of view, while concentrated inliers can be a repeated-texture artefact. The heritage setting plausibly inverts that preference many archive photographs share only a small unchanged feature with their modern counterpart but the choice deserves to be settled by a labelled subset rather than by intuition. This extension would not change the matcher’s algorithmic core; it would only replace the line in `compute_ransac_inliers` that returns `int(inlier_mask.sum())` with a small structured object carrying both the count and the spatial statistics of the inlier set.

CHAPTER 5

CONCLUSION

5.1 SUMMARY

This bachelor semester project delivered a per-photo geolocation refinement layer for the *Spatial Life of Images* programme: a pipeline that consumes `is_building = true` photographs from the upstream filtering stage, matches each one against modern Mapillary panoramas drawn from a 300 m search box around its Flickr coordinate, and outputs a verified match (a Mapillary image identifier, a predicted coordinate, and a bounded confidence score) or an explicit no-result decision.

The matching core combines three components developed in this project. A two-helper candidate sampler (`smart_angle_candidates` for a structured 3×3 grid with compass-angle binning, and `MapillarySampler` for a Gaussian-weighted bounding-box draw) returns a small viewpoint-diverse pool of Mapillary candidates per query. LoFTR pairs each (query, candidate) and returns pixel-wise correspondences. USAC_MAGSAC geometric verification at $\tau = 3.0$ pixels filters those correspondences down to a geometrically consistent inlier count, which is the matcher’s confidence signal both for ranking candidates and for the bounded p_{match} score persisted to the database.

The matcher is exposed as a single function, `find_matches`, whose output columns are similar to the schema of the shared `machine_learning_photo` table. The same function is consumed unchanged by a local CSV runner used during development and by Nathanael Lambert’s production database trainer through `pipeline.fast_mapillary`. The integration is operational: a row written into the database by the local path and a row written by the production path are indistinguishable.

APPENDIX A

PER-INSTITUTION BREAKDOWN

The table below lists the full per-institution breakdown of the 1,768-photograph working batch reported in Chapter 3. The top nine institutions are discussed with median statistics in §3.4; this table extends that summary to the long tail of smaller contributors.

TABLE A.1

All institutions contributing to the 1,768-photograph working batch, ranked by input count. “Matched” counts rows with `status = ok`; “Strong” counts rows with ≥ 15 RANSAC inliers; “Retained” counts rows with $p_{\text{match}} > 0.05$.

Institution	Input	Matched	Strong	Retained
Cornell University Library	413	367	68	8
Community Archives of Belleville & Hastings County	391	313	64	13
Swedish National Heritage Board	215	200	122	53
National Library of Ireland on The Commons	176	141	76	19
State Library of Queensland Australia	163	148	29	5
Powerhouse Museum Collection	65	65	34	6
Deseronto Archives	43	33	3	2
US National Archives	34	27	12	4
State Library and Archives of Florida	33	31	14	3
The National Archives UK	27	17	1	0
East Riding Archives	24	15	2	0
National Library of Norway	22	7	1	0
Médiathèques Valence Romans agglomération	18	14	2	0
National Library of Scotland	17	17	10	0
State Library of NSW	16	16	8	1
National Library NZ on The Commons	14	13	8	4
National Science and Media Museum	11	10	3	1
LIGC & NLW	11	9	3	1
New York Public Library	9	4	0	0
Australian National Maritime Museum on The Commons	7	6	3	1

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Institution	Input	Matched	Strong	Retained
Svenska litteratursällskapet i Finland	6	6	3	1
Bibliothèque de Toulouse	6	5	2	2
Smithsonian Institution	5	5	4	0
National Library of Australia Commons	5	4	0	0
Galt Museum & Archives on The Commons	4	0	0	0
Liberas	4	4	4	2
CADL Local History	3	3	1	0
Nationaal Archief	3	3	0	0
North East Museums	3	3	2	0
MHNSW – State Archives Collection	3	3	1	1
Museum of Photographic Arts Collections	3	2	0	0
The Library of Congress	2	2	1	0
Museum of Hartlepool	2	2	1	0
National Galleries of Scotland Commons	2	2	0	0
Stockholm Transport Museum Commons	2	2	2	0
Congregation of Sisters of St Joseph in Canada	1	1	0	0
National Archives of Estonia	1	1	0	0
NavyMedicine	1	1	0	0
Randolph Historical VT	1	1	0	0
Public Record Office of Northern Ireland	1	1	1	0
UA Archives & Upper Arlington History	1	0	0	0
Total (41 institutions)	1,768	1,504	485	127

APPENDIX B

CODE AND DATA AVAILABILITY

REPOSITORY

Source code, README, and requirements are available at:

`https://github.com/kimoal276/flickr-project`

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